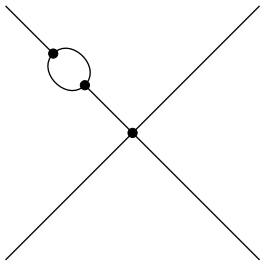
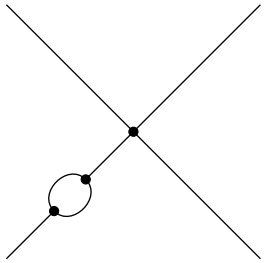
External contact bubbles: one selected leg

Four-point contact plus two trivalent insertion vertices. Plain diagnostics; no caps or circles.

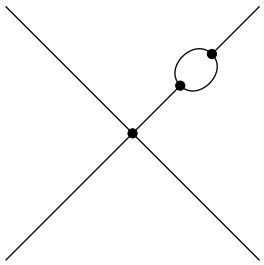
Slot 1: upper left



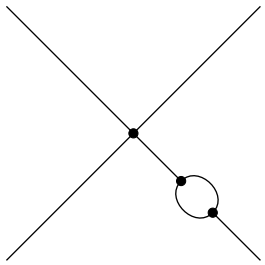
Slot 2: lower left



Slot 3: upper right



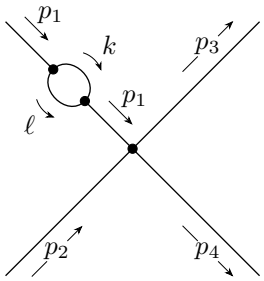
Slot 4: lower right



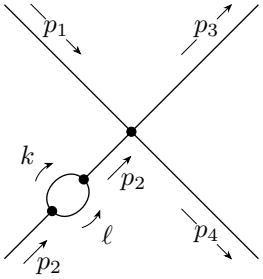
Logical momentum labels and mixed lines

Only the selected host leg is split; bubble arcs retain their own labels.

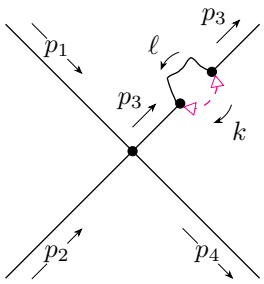
Slot 1



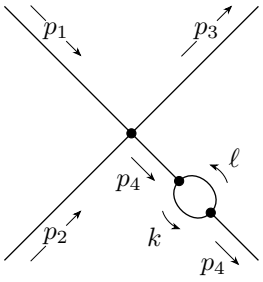
Slot 2



Slot 3: EW and Proca



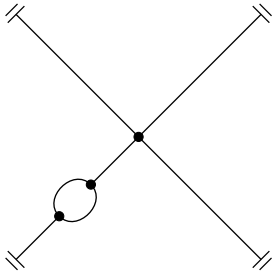
Slot 4: reversed loop



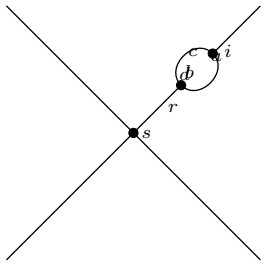
Separate controls and compatibility

Caps and indices appear only in their dedicated checks. Index placement is deferred (D01).

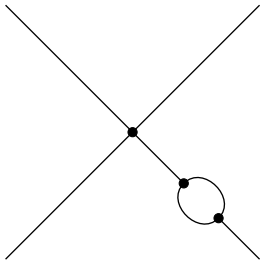
Slot 2: outer caps only



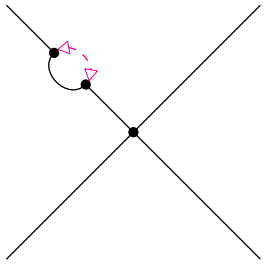
Slot 3: independent indices



Slot 4: shifted insertion



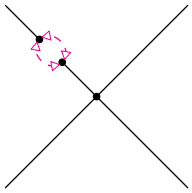
Slot 1: mixed bubble



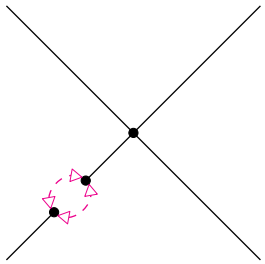
Proca endpoint contact on all four placements

Only the two bubble arcs are pink Proca lines. No momentum, caps or circles.

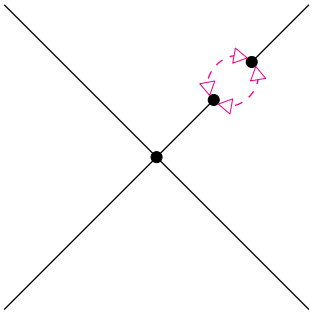
Slot 1; scale 0.72



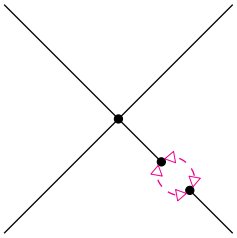
Slot 2; scale 1.0



Slot 3; scale 1.2



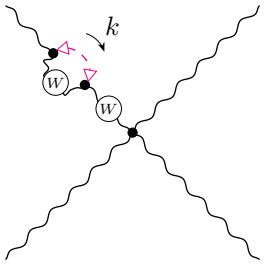
Slot 4; scale 0.9



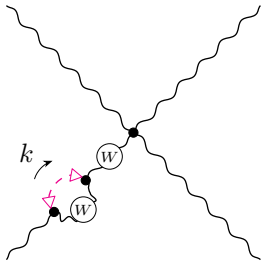
Species sample: EW W and pink scalar polarization

Plain EW external legs; inner host segment W; two-ended pink scalar loop arc, no circle; its momentum is k. Other loop arc is EW W with no momentum.

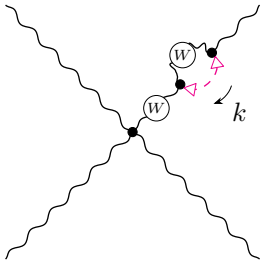
Slot 1



Slot 2



Slot 3



Slot 4

